

Dynamic Guideline: WHO-Based HIV - Diagnosis and/or Treatment and/or Management

I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

1.1 Guideline Overview

This report synthesizes **exactly five WHO HIV-related guideline documents or guideline updates** available in the supplied evidence base, with emphasis on diagnosis, treatment, service delivery, prevention, and management of advanced HIV disease.

No.	WHO Guideline / Guideline Update	Publication / Update Date Identified	Target Population	Scope and Objectives
1	WHO guidelines on the management of advanced HIV disease	18 December 2025	Adults, adolescents, and children living with HIV, especially those with advanced HIV disease	Identification and management of advanced HIV disease; CD4 testing; clinical staging; opportunistic infection package; post-discharge interventions; Kaposi sarcoma treatment; rapid ART and adherence support. WHO states this 2025 guideline consolidates relevant WHO guidance from 2021–2025 and adds new 2025 recommendations WHO HIV guidelines page , WHO AHD guideline PDF .
2	WHO updated recommendations on HIV clinical management: recommendations for a public health approach	2025; press-release republication dated 5 January 2026; WHO HIV page also lists a late-2025 update	People living with HIV, including adults, adolescents, pregnant/breastfeeding women, infants and children	Updated recommendations on ART optimization, vertical transmission management, breastfeeding, infant prophylaxis, and TB preventive treatment; intended for integration into consolidated WHO HIV guidelines NCBI Bookshelf executive summary , EATG/WHO press release .

No.	WHO Guideline / Guideline Update	Publication / Update Date Identified	Target Population	Scope and Objectives
3	Updated WHO guideline on HIV service delivery	12 September 2025	People living with HIV receiving long-term care; HIV programmes and health systems	Integrated, person-centred HIV service delivery; long-term ART adherence; retention; differentiated care; health and well-being of people living with HIV WHO service delivery update .
4	Guidelines on lenacapavir for HIV prevention and testing strategies for long-acting injectable pre-exposure prophylaxis	14 July 2025	People at substantial risk of HIV acquisition; PrEP users and programmes	Long-acting PrEP with lenacapavir; testing strategies for long-acting injectable PrEP; resistance considerations and linkage to ART if breakthrough infection occurs WHO HIV guidelines page , WHO HIV drug resistance fact sheet .
5	Consolidated guidelines on HIV prevention, testing, treatment, service delivery and monitoring	2021	All people affected by HIV across ages, populations and settings	WHO’s consolidated HIV care-continuum guideline covering prevention, testing, ART regimens, treatment monitoring, service delivery, advanced HIV disease, comorbidities, cervical precancer screening, and palliative care WHO consolidated HIV guidelines .

Related WHO youth-engagement item identified: The exact WHO title found in the supplied evidence is **“Meaningful youth engagement in a WHO guideline development process: case study of WHO abortion care guideline (2022)”**, launched through a WHO webinar on **11 December 2025**. It is **not HIV-specific**, but it is relevant to guideline-development equity, inclusivity and participation principles [WHO webinar](#). HIV-related youth-engagement evidence supports adolescent-friendly HIV services and meaningful youth participation in HIV programming, but no HIV-specific WHO guideline-development case study with that exact title was identified in the supplied sources [WHO Africa: HIV/AIDS](#), [PMC article](#).

1.2 Key Recommendations

GRADE interpretation used in this report

GRADE Category	Meaning for Clinical Confidence
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GRADE Category	Meaning for Clinical Confidence
High	Further research is very unlikely to change confidence in the estimate of effect.
Moderate	Further research may affect confidence and may change the estimate.
Low	Further research is likely to affect confidence and may change the estimate.
Very Low	Effect estimate is very uncertain.

Where WHO recommendation strength or certainty was explicitly reported in the supplied evidence, it is reproduced. Where detailed GRADE tables were not extractable from the supplied source snippets, the table labels the recommendation using the best available reported certainty or clearly notes limited extractability.

Recommendation	Population / Setting	GRADE Evidence Quality	Strength
Use CD4 testing as the preferred method to identify advanced HIV disease in people living with HIV.	Adults, adolescents, and children living with HIV where CD4 testing is available	Moderate	Strong
Where CD4 testing is not available, use WHO clinical staging to identify advanced HIV disease.	Settings with unavailable or inaccessible CD4 testing	Very Low	Conditional
Provide all people presenting with advanced HIV disease a package of care including screening, treatment and/or prophylaxis for major opportunistic infections, rapid ART initiation, and intensified adherence support.	Everyone presenting with advanced HIV disease	Moderate	Strong
Prioritize rapid ART initiation, defined by WHO as starting ART within seven days of HIV diagnosis, preferably same day when clinically appropriate.	People newly diagnosed with HIV, including those with advanced HIV disease after assessment for contraindications or OIs	Moderate	Strong
Provide intensified adherence support, including enhanced counselling, home visits or phone calls, and home-based or palliative care where needed.	People with advanced HIV disease, people returning to care, and people at risk of poor retention	Moderate	Strong

Recommendation	Population / Setting	GRADE Evidence Quality	Strength
Offer post-discharge support interventions for people with HIV hospitalized with serious illness to improve transition from hospital to outpatient care.	People with HIV after hospitalization for serious illness	Low	Conditional
For HIV-associated Kaposi sarcoma, use ART for all people with HIV and consider paclitaxel or pegylated liposomal doxorubicin as chemotherapy options when Kaposi sarcoma is confirmed.	People living with HIV and Kaposi sarcoma	Low for paclitaxel/pegylated liposomal doxorubicin summary; Very Low for some comparative chemotherapy evidence	Conditional
Confirm Kaposi sarcoma histopathologically before chemotherapy when feasible, because clinical diagnosis alone has suboptimal positive predictive value.	People with suspected Kaposi sarcoma	Low	Conditional
Use dolutegravir-containing regimens as preferred initial ART and for people failing NNRTI-based regimens.	ART-naïve people and people failing NNRTI-based regimens	Moderate	Strong
For people failing dolutegravir-containing regimens, use a protease inhibitor anchor drug; darunavir/ritonavir is now the preferred PI option.	People with treatment failure on dolutegravir-containing ART	Low to Moderate	Conditional
TDF or TAF can be recycled in subsequent ART regimens in adults and adolescents.	Adults and adolescents requiring subsequent ART regimens	Low to Moderate	Conditional
Support informed breastfeeding choices for mothers with HIV on ART; WHO continues to support exclusive breastfeeding for six months, continued breastfeeding to 12 months, and possibly to 24 months or longer with complementary feeding.	Mothers with HIV and infants	Moderate	Strong

Recommendation	Population / Setting	GRADE Evidence Quality	Strength
Use shorter rifapentine-based TB preventive treatment options, including 3HP, for eligible people living with HIV.	Eligible adults and adolescents living with HIV	Moderate	Strong
Use differentiated, integrated HIV service delivery models to support adherence and retention, including reduced visit frequency, multi-month dispensing, and community ART distribution where appropriate.	People established on ART and HIV programmes	Moderate	Strong
Use long-acting PrEP options, including lenacapavir and cabotegravir, with appropriate HIV testing strategies and resistance monitoring.	People at substantial risk of HIV acquisition	High to Moderate	Strong

Core WHO definition of advanced HIV disease: WHO defines advanced HIV disease in adults and adolescents as **CD4 count below 200 cells/mm³ or WHO clinical stage 3 or 4**; all children younger than five years are considered to have advanced HIV disease because of their higher risk of progression and mortality [WHO AHD page](#). CD4 testing is no longer required to start ART, but remains essential to identify advanced HIV disease [WHO AHD page](#).

1.3 Guideline Development Methodology

Evidence Review Process

WHO’s recent HIV guidance uses an evidence-informed public-health approach. The **2025 advanced HIV disease guideline** consolidates WHO recommendations published from 2021–2025 and adds new 2025 recommendations on CD4 testing, WHO clinical staging, post-discharge interventions, and Kaposi sarcoma management [WHO AHD guideline PDF](#). The guideline’s objectives include evidence-informed clinical recommendations for screening, prophylaxis, treatment, and management of opportunistic infections in people with advanced HIV disease, and implementation of the WHO-recommended AHD package of care [NCBI Bookshelf: Objectives](#).

Key evidence elements identified in the supplied sources include:

- A systematic review on diagnostic accuracy of WHO clinical staging for immunologically defined advanced HIV disease, cited by WHO as showing low accuracy with meaningful false-positive and false-negative risks [WHO AHD guideline PDF](#).
- Evidence supporting CD4 testing as the preferred identification method for advanced HIV disease, resulting in a new 2025 **strong recommendation** with **moderate-certainty evidence** [WHO AHD guideline PDF](#), [NCBI Bookshelf: CD4 testing and WHO clinical staging](#).

- Systematic review evidence for HIV-associated Kaposi sarcoma, including five randomized trials, although meta-analysis was not possible because of heterogeneous reporting [NCBI Bookshelf: Kaposi's sarcoma](#).
- ART, vertical transmission and TB prevention evidence summarized in WHO's updated clinical-management recommendations [NCBI Bookshelf executive summary](#).

Consensus Method

WHO convened guideline development processes using evidence review and Guideline Development Group deliberation. For advanced HIV disease terminology, a 2024 consensus process involved more than 900 participants; 47% were from the African Region, nearly half represented civil society or frontline workers, and 44% identified as living with HIV [WHO AHD guideline PDF](#). The consensus supported use of the term **advanced HIV disease** clinically to differentiate levels of care and communicate disease status, while recognizing that the older term **AIDS** remains relevant for advocacy and education [WHO AHD guideline PDF](#).

Conflict of Interest Management

The supplied source summaries do not provide detailed declarations-of-interest tables. WHO guideline development normally includes declaration and management of conflicts of interest; however, specific individual declarations for the five documents were not extractable from the supplied evidence snippets. This is an evidence-documentation limitation rather than evidence of absent conflict management.

II. Evidence Summary After WHO Latest Guideline Release (WHO guidelines on the management of advanced HIV disease) (Evidence Cutoff Date: 29 April 2026)

2.1 New Evidence Overview

Search Strategy

Evidence was reviewed from the supplied source summaries for the period after release of the WHO advanced HIV disease guideline on **18 December 2025** through **29 April 2026**, with contextual inclusion of WHO guidance and key sources published shortly before the release when they were incorporated into or directly relevant to WHO's 2025 recommendations.

Databases and Sources Searched

The supplied evidence base included:

- WHO official guideline pages and WHO IRIS PDFs.
- NCBI Bookshelf versions of WHO guideline chapters.
- WHO programme pages and advisory reports.
- Other authoritative guidance sources, including U.S. HHS/NIH and CDC guideline portals.
- Country guideline updates, including South Africa's February 2026 primary healthcare HIV chapter.
- Peer-reviewed or indexed sources where surfaced, including systematic reviews cited by WHO and a Frontiers in Medicine 2025 case report/literature review.
- Secondary clinical summaries of WHO updates, including Medthority and Positively Aware.

- Implementation summaries such as CQUIN materials.

Inclusion Criteria

Included evidence addressed one or more of the following:

1. HIV diagnosis or testing.
2. ART initiation, optimization, treatment failure, or monitoring.
3. Advanced HIV disease identification and management.
4. Opportunistic infection screening, prophylaxis, or treatment.
5. TB prevention or diagnosis in people living with HIV.
6. Cryptococcal disease screening and management.
7. Kaposi sarcoma treatment in people living with HIV.
8. HIV service delivery, adherence, retention, differentiated care.
9. HIV prevention with long-acting PrEP.
10. Youth engagement where relevant to HIV services or WHO guideline development.

Exclusion Criteria

Excluded or downgraded evidence included:

- Non-HIV WHO youth-engagement case studies except as guideline-methodology context.
- Secondary commentaries without clinical detail unless they summarized WHO recommendations.
- Studies without extractable design, population, or outcomes.
- Evidence unrelated to diagnosis, treatment, management, prevention, or service delivery for HIV.

2.2 Key New Studies

Study / Source	Design	Key Findings	GRADE Quality
WHO guidelines on the management of advanced HIV disease, 2025	WHO guideline using evidence review and GDG consensus	New 2025 recommendations: CD4 testing preferred for identifying AHD; WHO clinical staging may be used where CD4 testing unavailable; post-discharge support interventions; updated Kaposi sarcoma options WHO AHD guideline PDF .	Moderate for CD4 testing; Very Low for clinical staging; Low for post-discharge and KS chemotherapy summaries
Twabi et al., 2025, diagnostic accuracy of WHO clinical staging for advanced HIV disease	Systematic review and meta-analysis cited by WHO	WHO judged clinical staging accuracy as low because of substantial false positives and false negatives; implementation should not substitute for CD4 testing when CD4 is available WHO AHD guideline PDF .	Very Low

Study / Source	Design	Key Findings	GRADE Quality
WHO Kaposi sarcoma evidence review in AHD guideline	Systematic review of randomized trials; five RCTs identified; no meta-analysis possible	Evidence supports updating HIV-associated Kaposi sarcoma pharmacologic options, including paclitaxel or pegylated liposomal doxorubicin; certainty limited by heterogeneity and very-low certainty for some comparisons NCBI Bookshelf: Kaposi's sarcoma .	Low to Very Low
WHO updated recommendations on HIV clinical management, 2025/2026	WHO guideline update	Dolutegravir-containing regimens remain preferred for initial treatment and after NNRTI failure; PI anchor after DTG failure, with darunavir/ritonavir preferred; TDF/TAF can be recycled in subsequent regimens; breastfeeding guidance and TB prevention updated NCBI Bookshelf executive summary .	Moderate overall; recommendation-specific GRADE not fully extractable
South Africa primary healthcare HIV chapter, February 2026	National clinical guideline update	Reinforces CD4 testing, CrAg testing when CD4 <200 cells/mm ³ , lumbar puncture if CrAg positive, cotrimoxazole prophylaxis when CD4 <200, and TB diagnostic testing including GeneXpert and urine-LAM when TB suspected South Africa NDoH PDF .	Moderate for CD4/CrAg/TB diagnostic strategy; Conditional implementation strength
Medthority summary of WHO 2025 AHD update, March 2026	Clinical news summary of WHO guideline	Reports CD4 as preferred AHD identification method; clinical staging as conditional due to very-low certainty sensitivity/specificity; post-discharge interventions as conditional based on low-certainty evidence; paclitaxel or pegylated liposomal doxorubicin for KS based on low-certainty evidence Medthority .	Moderate for CD4; Very Low for clinical staging; Low for post-discharge and KS
CQUIN implementation summary, 2025	Implementation-oriented summary of WHO AHD recommendations	Highlights operational needs: point-of-care CD4 testing, training, task-sharing, QA, referral pathways, SOPs, post-discharge transitional care, peer counsellors, volunteers, nutritional support, and KS histopathology/chemotherapy access CQUIN PDF .	Low for implementation outcomes; Conditional programme strength

Study / Source	Design	Key Findings	GRADE Quality
WHO HIV service delivery update, September 2025	WHO guideline update	Supports integrated HIV service delivery, long-term adherence, retention, person-centred care, and differentiated delivery WHO service delivery update .	Moderate
WHO lenacapavir PrEP guidance and HIV drug-resistance summary, 2025	WHO prevention guideline and technical fact sheet	Lenacapavir is a six-monthly long-acting PrEP option; WHO identifies cabotegravir and lenacapavir as important PrEP advances; lenacapavir has no predicted cross-resistance to current WHO-recommended ART if breakthrough infection occurs WHO HIV drug resistance fact sheet .	High to Moderate
WHO youth-engagement evidence brief launch, 2025	WHO guideline-methodology case study; not HIV-specific	Case study concerns WHO abortion care guideline, not HIV; relevant principles include inclusivity, equity, lived experience, and meaningful youth participation in guideline development WHO webinar .	Low for direct HIV applicability; Conditional relevance

2.3 Evidence Gaps and Future Research Needs

1. Advanced HIV disease identification

- CD4 testing is strongly supported as the preferred method, but implementation research is needed on point-of-care CD4 scale-up, quality assurance, cost-effectiveness, and integration into primary care.
- WHO clinical staging has very-low certainty diagnostic accuracy and may miss substantial numbers of AHD cases; research is needed on combined clinical/laboratory triage tools.

2. Post-discharge care

- WHO’s post-discharge recommendation is conditional with low-certainty evidence. Pragmatic trials are needed to define the optimal package: early outpatient review, peer navigation, nutritional support, home visits, transport support, digital reminders, and community ART linkage.

3. Kaposi sarcoma

- Evidence for paclitaxel or pegylated liposomal doxorubicin in HIV-associated Kaposi sarcoma is limited by low or very-low certainty evidence and heterogeneous trial reporting.
- Future trials should compare chemotherapy regimens with standardized staging, response definitions, survival endpoints, toxicity reporting, and ART status.
- Better validation is needed for paediatric Kaposi sarcoma staging systems [NCBI Bookshelf: Kaposi’s sarcoma](#).

4. TB and cryptococcal disease in AHD

- More implementation research is needed on combined TB testing strategies using low-complexity NAATs, urine LF-LAM, chest radiography and specimen-specific testing in primary-care and inpatient settings.
- CD4-triggered cryptococcal antigen screening is operationally clear, but outcome studies are needed in diverse settings using reflex CrAg workflows and rapid linkage to lumbar puncture and antifungal treatment.

5. ART sequencing after dolutegravir failure

- WHO now supports PI-based subsequent regimens with darunavir/ritonavir preferred, but more real-world and trial data are needed on resistance, adherence barriers, tolerability, drug–drug interactions, cost, and outcomes in low- and middle-income countries.

6. Long-acting prevention and treatment

- Lenacapavir and cabotegravir are major prevention advances. Future evidence needs include implementation in adolescents, pregnant and breastfeeding populations, key populations, drug-resistance monitoring, and linkage after breakthrough infection.

7. Adolescents and meaningful youth engagement

- WHO and HIV programme sources support adolescent-friendly services and youth engagement, but no HIV-specific WHO guideline-development case study with the exact youth-engagement title was identified.
- Future WHO HIV guideline processes should document how young people, adolescents living with HIV, and young key populations are involved in scoping, evidence interpretation, recommendation formulation, implementation planning, and monitoring.

2.4 Updated Recommendations Based on New Evidence

Original Recommendation	New Evidence Impact	Updated Recommendation
Identify AHD using CD4 count or WHO stage 3/4 disease.	WHO 2025 now explicitly makes CD4 testing the preferred method ; clinical staging is less accurate and conditional where CD4 unavailable.	Use CD4 testing as the preferred AHD identification method for all people living with HIV where available. Use WHO clinical staging only as a fallback when CD4 testing is unavailable.
CD4 is not required before ART initiation.	Still true, but CD4 remains essential for AHD risk stratification and OI package activation.	Do not delay ART solely for CD4 testing, but obtain CD4 promptly at entry/re-entry to care to identify AHD and trigger OI screening/prophylaxis.

Original Recommendation	New Evidence Impact	Updated Recommendation
Provide AHD package of care.	WHO continues strong support; post-discharge care and KS updates strengthen the package.	For all AHD patients, implement a standardized package: CD4-based identification, TB testing, CrAg screening where indicated, cotrimoxazole prophylaxis when eligible, OI treatment/prophylaxis, rapid ART, adherence support, and planned follow-up.
Use clinical staging to detect advanced disease in low-resource settings.	Diagnostic accuracy evidence is very low; false negatives and false positives are clinically important.	Clinical staging should not replace CD4 testing where CD4 is available. If used, combine with referral pathways, staff training, and CD4 scale-up plans.
Routine discharge after stabilization.	WHO 2025 adds conditional support for post-discharge interventions due to high readmission and mortality risk.	Before discharge, create a transition plan: medical review date, ART/OI medication supply, adherence support, linkage to outpatient HIV care, peer/community support, nutrition assessment, and clear emergency return instructions.
ART is first-line for HIV-associated Kaposi sarcoma; chemotherapy options varied.	WHO 2025 updates KS pharmacologic guidance; paclitaxel or pegylated liposomal doxorubicin are suggested options.	Start or optimize ART in all people with HIV-associated KS. Confirm diagnosis histopathologically where feasible; for advanced/symptomatic KS, use paclitaxel or pegylated liposomal doxorubicin where available and appropriate.
Dolutegravir-based ART is preferred first-line therapy.	WHO clinical-management update confirms continued preference for DTG regimens.	Continue DTG-containing ART as preferred initial therapy and for NNRTI regimen failure unless contraindicated.
PI-based regimens are used after integrase-inhibitor failure.	WHO now identifies darunavir/ritonavir as preferred PI option after DTG failure.	After confirmed DTG-regimen failure, assess adherence and resistance where available; use a PI-anchored regimen, preferably darunavir/ritonavir, with optimized NRTI backbone.
TB preventive treatment should be offered to eligible people living with HIV.	WHO clinical-management update emphasizes shorter rifapentine-based regimens including 3HP.	Offer eligible people living with HIV shorter TPT options, including 3HP, with attention to drug interactions, pregnancy status, age, and local TB guidance.

Original Recommendation	New Evidence Impact	Updated Recommendation
HIV services should support adherence and retention.	WHO 2025 service-delivery guidance reinforces integrated, person-centred, differentiated delivery.	Use differentiated service delivery: multi-month dispensing, reduced visit frequency, community ART distribution, integrated comorbidity care, and targeted adherence support for unstable or high-risk patients.
PrEP options include oral and long-acting approaches.	WHO 2025 lenacapavir guidance and drug-resistance information support long-acting PrEP expansion.	Offer choice-based PrEP, including long-acting cabotegravir or six-monthly lenacapavir where available, with robust HIV testing before and during use and rapid linkage to ART after breakthrough infection.

Practical Clinical Algorithm: WHO-Based HIV and Advanced HIV Disease Management

1. At HIV diagnosis or re-entry to care

- Confirm HIV status using nationally validated WHO-aligned testing algorithms.
- Start ART rapidly, ideally same day or within seven days, unless evaluation identifies conditions requiring ART timing adjustment.
- Obtain baseline CD4 where available, not to delay ART but to identify AHD.

2. Identify advanced HIV disease

- AHD if CD4 <200 cells/mm³ or WHO stage 3/4 disease in adults/adolescents.
- All children <5 years presenting to care should be considered to have AHD unless clinically stable on ART for >1 year.
- If CD4 unavailable, use WHO clinical staging as a temporary fallback.

3. Activate AHD package

- Screen for TB, including symptom screen and appropriate diagnostic testing.
- Use urine LF-LAM and molecular tests where indicated and available.
- Screen for cryptococcal antigen when CD4 thresholds indicate, especially CD4 <200 cells/mm³ in some national algorithms.
- Provide cotrimoxazole prophylaxis when eligible.
- Treat or prevent major OIs.
- Provide rapid ART with careful timing in TB meningitis or cryptococcal meningitis.
- Intensify adherence and retention support.

4. ART selection

- Use dolutegravir-containing regimens as preferred initial ART.
- After NNRTI failure, use DTG-based regimens where appropriate.
- After DTG failure, use PI-anchored subsequent therapy; darunavir/ritonavir is preferred in WHO's updated clinical-management recommendations.

5. Hospital discharge for serious illness

- Do not discharge with only a prescription.
- Provide transition plan, OI medication supply, ART plan, follow-up appointment, adherence support, referral pathway, nutrition/social support, and community linkage.

6. Kaposi sarcoma

- Start or continue ART in all.
- Confirm diagnosis histopathologically when feasible.
- For advanced or symptomatic disease, consider paclitaxel or pegylated liposomal doxorubicin, depending on availability, contraindications, and oncology capacity.

7. Long-term service delivery

- For stable patients: differentiated care, multi-month dispensing, reduced clinic visits, community ART options.
- For unstable, newly diagnosed, AHD, pregnant/breastfeeding, paediatric, adolescent, or returning-to-care patients: enhanced follow-up and tailored support.

Summary Conclusion

As of **29 April 2026**, WHO-based HIV management emphasizes **rapid ART for all, dolutegravir-centered treatment, differentiated and integrated service delivery, choice-based prevention including long-acting PrEP**, and a strengthened approach to **advanced HIV disease**. The most consequential 2025 AHD shift is the strong recommendation that **CD4 testing is the preferred method for identifying advanced HIV disease**, while WHO clinical staging is only a conditional fallback where CD4 testing is unavailable. For patients with AHD, ART alone is insufficient: programmes should deliver a complete package including OI screening, prophylaxis and treatment, rapid ART, adherence support, post-discharge care, and updated Kaposi sarcoma management.